Analytical Chemistry Cumulative Exam
January 24, 2014

Topic: Problems you should be able to solve from the Analytical Chemistry I course.

1) Using propagation of error correctly, calculate the molarity of a caffeine standard made by weighing out 0.1941 grams of caffeine on a balance that has a known absolute error of 0.0002 grams. The caffeine was then transferred to a 100.00 ± 0.08 mL volumetric flask and dissolved in water. (The molecular weight for caffeine is 194.1 g/mol.)

2) Describe how you would make a 1.0 L sodium acetate buffer, pH 5.0 with an ionic strength (μ) = 0.2 M. The pK_a for acetic acid is 4.75.

3) Find the solubility of PbF_2 ($K_{sp} = 3.6 \times 10^{-8}$) in a solution with the pH fixed to 5.0. The K_a for HF is 6.8×10^{-4}.

4) For oxygen, consider its standard reduction potential:

$O_2 + 4H^+ + 4e^- \leftrightarrows 2H_2O$ $E°$ +1.229 V

What is the value of $E°'$ at pH 7?

List of Equations.

$$e_y = \sqrt{e_{x1}^{\ 2} + e_{x_2}^{\ 2} + \ldots}$$

$$\%e_y = \sqrt{\%e_{x1}^{\ 2} + \%e_{x_2}^{\ 2} + \ldots}$$

$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$$

$$z = \frac{x - \mu}{\sigma}$$

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{n}}$$

$$t_{calculated} = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{pooled}} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

$$s_{pooled} = \sqrt{\frac{s_1^2(n_1 - 1) + s_2^2(n_2 - 1)}{n_1 + n_2 - 2}}$$

$$s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}}$$

$$t_{calculated} = \frac{|\bar{d}|}{s_d}\sqrt{n}$$

$$F_{calculated} = \frac{s_1^2}{s_2^2}$$

$$\mu = \frac{1}{2}\sum c_i z_i^{\ 2}$$

$$-\log \gamma_i = 0.5 z_i^{\ 2}\sqrt{\mu}$$

$$pH = \frac{1}{2}(pK_{a1} + pK_{a2})$$

$$\alpha(H_2A) = \frac{[H^+]^2}{[H^+]^2 + K_1[H^+] + K_1K_2}$$

$$\alpha(HA^-) = \frac{K_1[H^+]}{[H^+]^2 + K_1[H^+] + K_1K_2}$$

$$\alpha(A^{2-}) = \frac{K_1K_2}{[H^+]^2 + K_1[H^+] + K_1K_2}$$

$$E = E^0 - \frac{0.059}{n}\log(Ratio)$$

$$E_{cell} = E_+ - E_-$$

$$R_s = \frac{\sqrt{N}}{4}\frac{\alpha - 1}{\alpha}\frac{k'}{1 + k'}$$